Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Small-Molecule Inhibitors Targeting YAP/TAZ–TEAD Signaling for Next-Generation Cancer Therapy			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio Medical		Bio Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Bio	Small-Molecule Pharmaceuticals	Industrial Biotechnology	Bio-Chemical Materials
Concept of Proposed Technology	The proposed technology focuses on the development of **first-in-class, TEAD1-selective small-molecule inhibitors** targeting aberrant **YAP/TAZ–TEAD transcriptional signaling**, a central oncogenic driver in multiple solid tumors. While the Hippo–YAP/TAZ–TEAD axis represents a compelling therapeutic target, prior clinical efforts have been limited by **dose-limiting toxicities associated with pan-TEAD inhibition**, which disrupts essential physiological transcriptional programs. To overcome these limitations, this project emphasizes **isoform-selective modulation of TEAD1**, leveraging **structure-based drug design (SBDD)**, rapid medicinal chemistry–driven structure–activity relationship (SAR) optimization, and **human-relevant three-dimensional (3D) cancer organoid models**. These organoid platforms capture transcriptional dynamics, tissue architecture, and context-dependent signaling behaviors that are not recapitulated in conventional two-dimensional assays. A defining feature of the proposed technology is the **closed-loop integration of medicinal chemistry with organoid-based functional validation**, incorporating quantitative phenotypic readouts such as viability, differentiation states, transcriptional programs, and other functional			

	signatures. This integrated discovery framework is designed to accelerate hit-to-lead optimization, identify safety liabilities early, and generate translationally robust data suitable for global clinical development and partnering.
Rationale for Joint R&D Project	Despite strong biochemical rationale, transcription factor–targeted oncology programs frequently fail due to insufficient functional validation in human-relevant systems. YAP/TAZ–TEAD signaling is highly context-dependent and influenced by cell–cell interactions, mechanical cues, and three-dimensional tissue organization, none of which are adequately modeled using standard 2D cell assays. The overseas research institute contributes **advanced expertise in organoid-based disease modeling, microphysiological systems, and quantitative functional phenotyping**, enabling direct evaluation of TEAD1-selective inhibition in clinically meaningful cancer contexts. These capabilities allow interrogation of pathway suppression, lineage plasticity, phenotypic efficacy, and early resistance mechanisms at a resolution not achievable with traditional models. The Korean SME partner contributes **agile medicinal chemistry, TEAD-focused intellectual property, and rapid SAR optimization capabilities**, enabling efficient translation of biological insights into differentiated chemical matter. The **chemistry–organoid co-optimization loop** formed through this collaboration is essential for reducing development risk, improving decision quality, and accelerating progression toward non-clinical readiness, making international joint R&D indispensable.
Key Words	1. TEAD1 selective inhibitor 2. YAP/TAZ-TEAD signaling 3. Small-molecule drug discovery 4. Organoid-based 3D disease modeling 5. Targeted oncology

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (2)	**End of the Project**	Level (4)
R&D Objectives	**Overall Objective** To establish a **TEAD1-selective anticancer drug discovery platform** and generate a **validated lead compound series** with human-relevant translational evidence suitable for IND-enabling studies and global partnering. **Specific Objectives** • Identify TEAD1-selective inhibitor scaffolds with clear biochemical and structural differentiation from pan-TEAD inhibitors • Demonstrate functional suppression of YAP/TAZ–TEAD transcriptional programs in human cancer organoids relevant to TEAD-driven malignancies • Establish mechanistic evidence that TEAD1-selective inhibition preserves anticancer efficacy while mitigating safety risks associated with pan-TEAD blockade • Generate a lead compound series with integrated biochemical, cellular, and organoid-based validation			

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R&D Target Specifications	The project targets the following quantitative and qualitative specifications: **1) Target Engagement and Selectivity** • TEAD1 biochemical inhibition IC_{50} ≤ 100 nM • ≥ 10-fold selectivity over TEAD2/3/4 confirmed by orthogonal assays **2) Cellular Functional Activity** • ≥ 70% inhibition of YAP/TAZ–TEAD transcriptional activity in relevant cell-based assays • Clear functional differentiation from pan-TEAD inhibition at the transcriptional level **3) Organoid-Based Validation** • Reproducible efficacy in YAP/TAZ-dependent human cancer organoid models • Confirmation of TEAD1 pathway suppression and disease-relevant phenotypic modulation **4) Drug-Like Properties** • cLogP ≤ 4.5 • Human liver microsome stability $t_1/_2$ ≥ 30 minutes • Acceptable solubility and chemical stability for progression **5) Translational Readiness** • Measurable pathway biomarkers supporting TEAD1-specific mechanism of action • Consolidated biochemical, cellular, and organoid data package suitable for non-clinical planning and global partnering **Final Output:** A TEAD1-selective lead compound series with reduced safety risk and validated human-relevant efficacy.
R&D Plan	The project will be executed through a **closed-loop discovery framework** integrating structure-based drug design, medicinal chemistry, and human organoid-based translational validation. **Phase 1: Hit Identification and TEAD1-Focused Design** *(Lead: Korean SME | Support: Research Institute)* • Structure-based virtual screening and rational scaffold selection targeting TEAD1 • Synthesis of initial hit compounds and biochemical evaluation of TEAD1 binding and selectivity • Early cellular reporter assays to confirm YAP/TAZ–TEAD pathway engagement **Milestones and Deliverables** • TEAD1-active hit compounds (IC_{50} ≤ 1 μM) • Preliminary TEAD isoform selectivity profiles • Initial SAR hypotheses --- **Phase 2: Lead Optimization through Medicinal Chemistry–Organoid Feedback**

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	(Joint Execution) • Iterative SAR optimization to improve potency, TEAD1 selectivity, and physicochemical balance • Parallel evaluation in advanced cancer organoid models assessing functional pathway suppression and phenotypic efficacy • Integration of organoid-derived data into compound prioritization **Milestones and Deliverables** • Optimized lead candidates (TEAD1 IC_{50} ≤ 100 nM; ≥ 10-fold isoform selectivity) • Correlated biochemical, cellular, and organoid activity datasets • Identification of response-associated transcriptional signatures --- **Phase 3: Translational Differentiation and Lead Series Nomination** *(Lead: Research Institute | Support: Korean SME)* • Comparative profiling versus pan-TEAD inhibitors to demonstrate functional differentiation • Mechanism-of-action confirmation through pathway biomarker and transcriptional analyses • Early assessment of resistance and compensatory signaling mechanisms **Milestones and Deliverables** • TEAD1-selective lead compound series with validated human-relevant efficacy • Translational biomarker panel supporting TEAD1-specific mechanism of action • Decision-grade, partner-ready data package

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will lead **human-relevant functional validation and translational characterization**, including: • Establishment and operation of disease-relevant 3D cancer organoid models driven by YAP/TAZ–TEAD signaling • Organoid-based pharmacology assessing dose–response, pathway suppression, and differentiation from pan-TEAD inhibitors • Mechanism-of-action studies using transcriptional profiling and pathway biomarker analysis • Identification of early resistance or compensatory signaling mechanisms • Generation of decision-grade datasets supporting medicinal chemistry prioritization and milestone review
Expected R&R of Korean SME Partner	The Korean SME partner will lead **compound discovery, optimization, and project execution**, including: • Design and synthesis of TEAD1-focused compound libraries using structure-based drug design • Biochemical and cellular assays to refine potency, selectivity, and target engagement • Early developability and ADME risk assessment • Integration of organoid-derived data into medicinal chemistry decision-making • Overall project coordination, milestone tracking, and intellectual property strategy

	• Preparation of partner-ready data packages for global licensing or co-development

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Scientific and Technological Impact** • Establishment of TEAD1 as a differentiated and clinically actionable anticancer target • Demonstration of isoform-selective transcriptional modulation with improved therapeutic index • Validation of an integrated chemistry–organoid discovery platform **Industrial and Commercial Impact** • Delivery of high-quality lead compound series suitable for non-clinical development • Increased attractiveness for global pharmaceutical partnering and licensing • Strengthening of Korean SME capabilities in advanced oncology drug discovery **Societal and Strategic Impact** • Contribution to safer and more effective targeted cancer therapies • Advancement of human-relevant drug discovery methodologies • Enhanced global competitiveness of Korea's biopharmaceutical innovation ecosystem

6. References

(1) Journal of Medicinal Chemistry, 2024. DOI:10.1021/acs.jmedchem.4c01393

(2) Bioorganic Chemistry, 2025. DOI: 10.1016/j.bioorg.2025.108425